Technical Design Document (TDD)

Moffat Bay Lodge Reservation System

Course: CSD460-300O Capstone in Software Development (2271-DD)
Project: Moffat Bay Lodge Reservation System
Client: San Juan Islands First Nations Development Committee
Location: Moffat Bay, Joviedsa Island, Washington State
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1. Introduction

1.1 Purpose

The purpose of the Moffat Bay Lodge Reservation System is to provide an online platform that enables customers to view lodge information, search for room availability, make reservations, manage bookings, and communicate with lodge staff. The system will streamline reservation management for administrators while improving customer experience through a user-friendly web interface.

The application will support:

- User account registration and authentication
- Lodge and room viewing
- Reservation booking and cancellation
- Availability checking
- Customer profile management
- Administrative reservation management
- Secure payment processing (future enhancement)
- Customer communication and support

Business Goals

- Increase reservation efficiency
- Reduce manual booking processes
- Improve customer satisfaction
- Minimize booking conflicts
- Maintain accurate reservation records
- Provide real-time room availability

Project Scope

Included Features

Public Website Features

- Marketing Landing Page
- About Us Page
- Contact Us Page
- Attractions Page

Customer Features

- Customer Registration
- Customer Login/Logout
- Session Management
- Reservation Booking
- Reservation Confirmation
- Reservation Lookup

Database Features

- Customer Management
- Reservation Storage
- Reservation Search
- User Authentication

Security Features

- Password Hashing
- Session Validation
- Input Validation
- Email Validation

Future Enhancements

The following features are outside the current project scope:

- Online payment processing
- Mobile application
- Rewards program
- Email notifications
- Reservation modification functionality
- Reservation cancellation after confirmation
- Administrative reporting dashboard
- Integration with travel websites

1.2 Terminology

Term	Definition
Customer	Registered user who can make reservations
Guest	Website visitor without an account
Reservation	Confirmed lodge booking
Reservation ID	Unique identifier assigned to each reservation
Customer ID	Unique identifier assigned to each customer
Authentication	Process of validating a customer's login credentials
Authorization	Process of determining access rights
Session	Temporary storage used to maintain login status
Check-In Date	Arrival date selected by the customer
Check-Out Date	Departure date selected by the customer
Room Type	Accommodation category selected by the customer
ERD	Entity Relationship Diagram
JDBC	Java Database Connectivity
Servlet	Java technology used to process requests
MySQL	Relational database system used for data storage
Password Hashing	Secure conversion of passwords into encrypted values
Agile Development	Iterative software development methodology
Kanban Board	Project management tool used to track tasks

User Story	Requirement described from a user's perspective
Story Point	Estimated effort required to complete a feature
TDD	Technical Design Document
QA Testing	Quality Assurance process used to verify functionality

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1.3 User Personas

Persona 1: Vacation Traveler

Name: Ruth Johnson

Age: 40

Occupation: Marketing Manager

Technical Skill Level: Intermediate

Goals:

- Plan family vacations online
- View room availability quickly
- Complete reservations without calling the lodge
- Receive booking confirmations immediately

Pain Points:

- Complicated booking systems
- Unclear pricing information
- Overbooking issues
- Slow website performance

Needs:

- Easy navigation
- Secure online reservations
- Instant confirmation emails
- Reservation modification options

Persona 2: Returning Customer

Name: Thomas Davis

Age: 52

Occupation: Business Owner

Technical Skill Level: Advanced

Goals:

- Quickly book favorite rooms
- Access booking history
- Modify reservations online
- Save personal information

Pain Points:

- Re-entering customer details
- Lengthy booking processes
- Lack of reservation history

Needs:

- User account access
- Fast reservation management
- Secure profile storage
- Reservation tracking

Persona 3: Lodge Administrator

Name: Anna Roberts

Age: 46

Occupation: Lodge Manager

Technical Skill Level: Intermediate

Goals:

- Manage reservations efficiently
- Monitor room occupancy
- Update room availability
- Serve customers effectively

Pain Points:

- Double bookings
- Difficult record management
- Manual reservation tracking

Needs:

- Administrative dashboard
- Reservation reports
- User management tools
- Real-time reservation updates

1.4 User Stories

High-Priority User Stories

US-001

Persona: Vacation Traveler

Priority: High

Story:

As a vacation traveler, I want to search for available rooms by date so that I can find accommodations that match my vacation schedule.

Acceptance Criteria:

- User enters check-in date
- User enters check-out date
- System displays available rooms
- System prevents unavailable room selection

Estimated Story Points: 8

US-002

Persona: Vacation Traveler

Priority: High

Story:

As a vacation traveler, I want to complete an online reservation so that I can secure my lodging before arrival.

Acceptance Criteria:

- Customer selects room
- Customer enters information
- Reservation is stored
- Confirmation number is generated

Estimated Story Points: 13

US-003

Persona: Lodge Administrator

Priority: High

Story:

As a lodge administrator, I want to manage reservations so that I can prevent conflicts and maintain accurate occupancy records.

Acceptance Criteria:

- Administrator views reservations
- Administrator edits reservations
- Administrator cancels reservations
- Updates are saved immediately

Estimated Story Points: 8

Medium Priority User Stories

US-004

Persona: Returning Customer

Priority: Medium

Story:

As a returning customer, I want to create an account so that my personal information can be saved for future reservations.

US-005

Persona: Returning Customer

Priority: Medium

Story:

As a returning customer, I want to log into my account so that I can access reservation history.

US-006

Persona: Returning Customer

Priority: Medium

Story:

As a returning customer, I want to modify an existing reservation so that I can adjust my travel plans when necessary.

Low Priority User Stories

US-007

Persona: Vacation Traveler

Priority: Low

Story:

As a vacation traveler, I want to view room details and amenities so that I can choose the best room.

US-008

Persona: Lodge Administrator

Priority: Low

Story:

As a lodge administrator, I want to generate occupancy reports so that I can monitor lodge performance.

US-009

Persona: Lodge Administrator

Priority: Low

Story:

As a lodge administrator, I want to manage customer accounts so that I can assist guests when account issues arise.

1.5 Work Estimations (To-Do List)

Breakdown of Top 3 User Stories

US-001: Search Available Rooms

Task	Hours
Create room search page UI	4
Design search form validation	2
Develop availability search logic	6
Connect database queries	3
Perform unit testing	3
Perform integration testing	2
Total Hours	**20**

US-002: Online Reservation

Task	Hours
Design reservation form	4
Create reservation database table	3
Develop booking functionality	8
Create confirmation page	3
Implement reservation validation	3

Test booking workflow	5
Total Hours	**26**

US-003: Reservation Administration

Task	Hours
Build admin dashboard	6
Create reservation management UI	5
Develop edit reservation functionality	4
Develop cancellation functionality	3
Security and access control implementation	3
Test administrator functions	3
Total Hours	**24**

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2. Design

2.1 Prototypes

To be completed in a later module.

2.2 ERD

To be completed in a later module.

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3. QA Testing

3.1 QA Test Plan

To be completed in a later module.

Additional Real-World Requirements Recommended for the Moffat Bay Project (Optional)

To make this TDD more professional and aligned with industry standards, we can include these considerations during future development:

Functional Requirements

- User registration and account recovery
- Email reservation confirmations
- Reservation modification and cancellation
- Room availability updates in real time
- User profile management
- Administrative reporting

Non-Functional Requirements

- Page load time under 3 seconds
- 99% application availability
- HTTPS encryption
- Mobile-responsive design
- WCAG accessibility compliance
- Cross-browser compatibility

Security Requirements

- Encrypted passwords
- Role-based access control
- Input sanitization
- Secure session handling
- Protection against SQL injection
- Audit logging for administrative actions

Future Enhancements

- Online payment processing
- Promotional discount codes
- Customer reviews and ratings
- SMS notifications
- Mobile application
- Business intelligence reporting